

Problem Set 1

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Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

```
1 #Q1, 1. Construct a 90% confidence interval for the mean IQ.
2 # CI = sample mean +/- critical value * SE
3
4 #step 1: For CI, must calculate the sample mean of IQs
5 y_samp_mean <- mean(y)
```

```

6 y_samp_mean #sample mean is 98.44
7
8 #step 2: then find the sample sd using the sample mean
9 y_sd <- sd(y)
10 y_sd #sd of y is 13.09287
11
12 #step 3: get n (sample size of students)
13 n <- length(y)
14 n #n = 25 -> a small sample
15
16 #step 4: find standard error (estimates the SD of the sampling
      distribution)
17 y_se <- y_sd / sqrt(n)
18 y_se #standard error = 2.618575
19
20 #step 5: find the critical t-value
21 # n is small, so sample sd is an uncertain estimate of the unknown
      population
22 # SD -> use the t distribution (thicker tails) with df = n - 1 = 24
23 # for a 90% CI I leave 5% in each tail, so need the value at p = 0.95
24 alpha <- 0.10
25 t_val <- qt(1 - alpha/2, df = n-1)
26 t_val #t value = 1.710882
27
28 #step 6: margin of error = critical t value * SE
29 margin_error <- t_val * y_se
30 margin_error #margin_error = 4.480072
31
32 #step 7: use margin of error to find the lower and higher ends of CI
33 lower_end_CI <- y_samp_mean - margin_error
34 upper_end_CI <- y_samp_mean + margin_error
35 lower_end_CI #lower end CI is 93.95993
36 upper_end_CI #upper end CI is 102.9201
37 #CI is [93.96, 102.92]
38
39 #check with the built-in function
40 t.test(y, conf.level = 0.90)
41 # ANSWER: 90% CI = [93.96, 102.92], same as by hand

```

Answer: The 90% confidence interval for the average student IQ is [93.96, 102.92].

A confidence interval is the sample mean plus or minus a margin of error, where the margin of error is a critical value times the standard error:

$$\bar{y} \pm t_{1-\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

I use the t-distribution rather than the normal (z) distribution mainly because the sample is small. With only 25 students, the sample standard deviation is an uncertain estimate of the population standard deviation, which is unknown. The t-distribution accounts for this extra uncertainty with thicker tails, which gives a slightly wider, more

cautious interval. With a large sample, the two distributions would give almost the same result.

The sample has $n = 25$, a mean of $\bar{y} = 98.44$, and a standard deviation of $s = 13.09$. For a 90% interval, $\alpha = 0.10$, so 5% is left in each tail, with $df = n - 1 = 24$:

$$SE = \frac{13.09}{\sqrt{25}} = 2.619 \quad t_{0.95, 24} = 1.711$$

$$98.44 \pm 1.711 \times 2.619 = 98.44 \pm 4.48 = [93.96, 102.92]$$

The built-in `t.test()` function gives the same interval.

This means I am 90% confident that the true mean IQ of students at this school lies between 93.96 and 102.92. In other words, if we repeatedly took random samples of 25 students and built an interval this way each time, about 90% of those intervals would contain the true mean. The national average of 100 falls inside this interval, so the sample is consistent with the school being about average.

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```

1 #Q1, 2. Test if school's mean IQ is greater than 100 (alpha=0.05)
2 # Hypotheses: one-sided, because the counselor asks whether the mean is
3 # higher than 100 (a specific direction), not just "different" from 100
4 # H0: average student IQ in the school is less than or equal to 100 (mu
  <= 100)
5 # HA: average student IQ in the school is greater than 100 (mu > 100)
6
7 #step 1: t statistic = (sample mean - 100) / SE, tested at 100 (edge of
  H0)
8 t_stat <- (y_samp_mean - 100) / y_se
9 t_stat #t_stat = -0.5957439
10
11 #step 2: find the p-val from the upper tail, since HA is mu > 100
12 # lower.tail = FALSE gives the probability of a t value at least this
  large
13 p_val <- pt(t_stat, df = n-1, lower.tail = FALSE)
14 p_val # p_val = 0.7215383
15
16 #check with the built-in function
17 t.test(y, mu = 100, alternative = "greater")
18
19 # ANSWER: p = 0.722 > 0.05 -> fail to reject H0. Not enough evidence that
  the
20 # school's mean IQ is greater than 100 (& the sample mean, 98.44, is
  below 100).
```

Answer: No. There is not enough evidence at $\alpha = 0.05$ to conclude that the school's average IQ is higher than 100.

I use a one-sided test because the counselor's question has a specific direction: whether the school's average is *higher* than 100, not just different from it. The hypotheses are:

$$H_0 : \mu \leq 100 \quad H_A : \mu > 100$$

I use a t-test for the same reason as in part 1: the sample is small and the population standard deviation has to be estimated from it.

The test statistic measures how many standard errors the sample mean is from 100:

$$t = \frac{\bar{y} - \mu_0}{SE} = \frac{98.44 - 100}{2.619} = -0.596$$

The test is calculated at $\mu_0 = 100$ because it is the edge of H_0 : if we cannot reject at 100, we cannot reject for any value below it either. Because H_A is $\mu > 100$, the p-value is the area in the upper tail of the t-distribution with 24 degrees of freedom (`lower.tail = FALSE`):

$$p = P(T_{24} \geq -0.596) = 0.722$$

The built-in `t.test()` function with `alternative = "greater"` gives the same result. Since $p = 0.722 > \alpha = 0.05$, I fail to reject the null hypothesis. This is not surprising. The sample mean (98.44) is actually below 100, so the data do not point in the direction of the alternative hypothesis at all.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

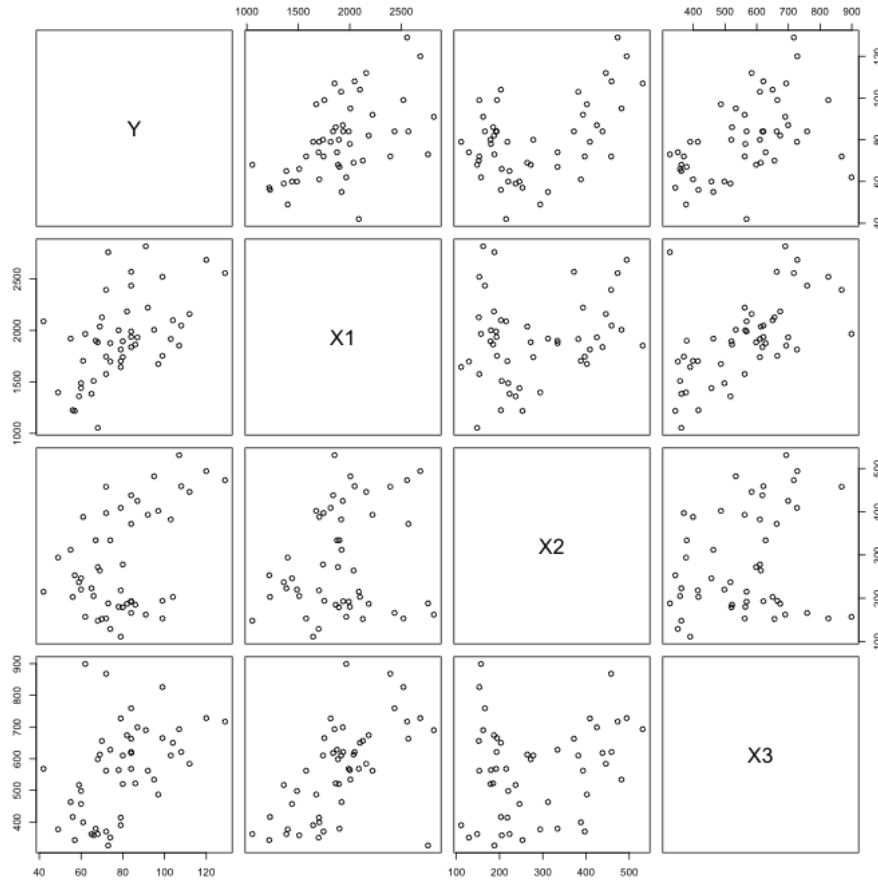
```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

```
1 #explore data
2 head(expenditure)
3 summary(expenditure)
4 str(expenditure)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```
1 #Q2, 1. Plot relationships among vars. What are the correlations?
2 #step 1: subset the dataset to only look at the variables I want
3 # plot() on a df of numeric columns makes a scatterplot matrix
4
5 png("graph-1.png", width = 700, height = 700)
6 expenditure %>%
7   select(Y, X1, X2, X3) %>%
8   plot()
9 dev.off()
10
11 #step 2: calculate the correlations among them
12 expenditure %>%
13   select(Y, X1, X2, X3) %>%
14   cor()
15 # All positive. Y-X1 = 0.532, Y-X2 = 0.448, Y-X3 = 0.464,
16 # X1-X2 = 0.206, X1-X3 = 0.595 (strongest), X2-X3 = 0.221
```

Figure 1: Scatterplot matrix of Y, X1, X2, and X3.



Answer: All four variables are positively related to each other, but none of the relationships are strong.

I use a scatterplot matrix because it shows every pair of variables at once, and `cor()` to measure each relationship with a correlation coefficient (r). As Figure 1 shows, per capita housing assistance (Y) has a moderate positive correlation with per capita income X1 ($r = 0.532$), the number of financially insecure residents X2 ($r = 0.448$), and the share of people in urban areas X3 ($r = 0.464$). States that score higher on each of these tend to spend more on housing assistance, although there is a lot of scatter around this.

Among the other variables, the strongest relationship is between X1 and X3 ($r = 0.595$): richer states also tend to be more urban. The relationships between X1 and X2 ($r = 0.206$) and between X2 and X3 ($r = 0.221$) are weak in comparison.

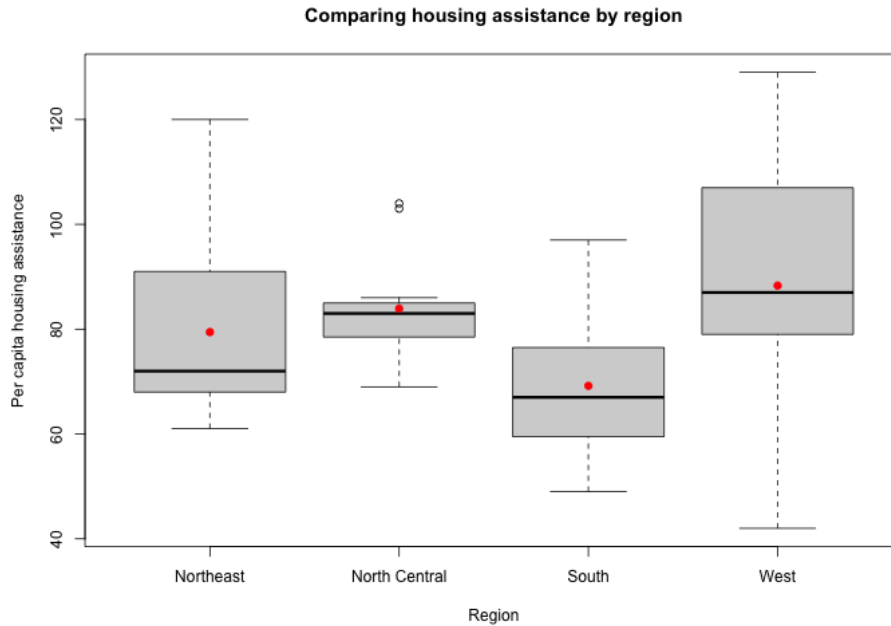
- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```

1 #Q2, 2. Plot relationship btwn Y and region. Which region has highest p/c
2 #expenditure on housing assistance
3
4 #step 1: make region (stored as numeric) a factor
5 class(expenditure$Region)
6 expenditure$Region <- factor(
7   expenditure$Region,
8   levels = c(1, 2, 3, 4),
9   labels = c("Northeast", "North Central", "South", "West")
10 )
11
12 #step 2: y is numeric, region is categorical —> use boxplot
13 png("graph_y_region.png", width = 700, height = 500)
14 plot(expenditure$Region, expenditure$Y,
15       main = "Comparing housing assistance by region",
16       xlab = "Region",
17       ylab = "Per capita housing assistance")
18 points(tapply(expenditure$Y, expenditure$Region, mean), col = "red", pch
19         = 19)
19 dev.off()
20
21 #The red dots show the mean of each region and the black lines show the
22 #median.
23 #The West has the highest median and mean (about 88.3) but also the
24 #widest
25 #spread, while the South has the lowest median and mean (about 69.2).
26 #Central's values are tightly clustered, with two high outliers. The
27 #Northeast's
28 #mean (79.4) is well above its median, so a few high-spending states are
29 #pulling its average up.
30
31 #Step 3: confirm means by region
32 expenditure %>%
33   group_by(Region) %>%
34   summarise(mean_y = mean(Y))
35 #mean findings: NE = 79.4, NC = 83.9, S = 69.2, W = 88.3
36 # The West has the highest average per capita expenditure on housing
37 # assistance

```

Figure 2: Boxplot of Y by Region (red dots show the mean of each region).



Answer: The West has the highest average per capita expenditure on housing assistance, with a mean of about 88.3.

I use a boxplot because Y is numeric and Region is categorical, so it compares the distribution of Y across the four regions side by side. Figure 2 shows that the South generally spends the least, while the West and North Central tend to spend more. The West also has the widest spread, while North Central's values are tightly clustered, with two high outliers.

Because a boxplot shows each region's median rather than its mean, I calculated the mean for each region and added it to the plot as a red dot. The mean per capita expenditure on housing assistance is approximately 79.4 in the Northeast, 83.9 in North Central, 69.2 in the South, and 88.3 in the West. Therefore, the West has the highest average per capita expenditure on housing assistance, with a mean of approximately 88.3. The Northeast's median is noticeably lower than its mean, which may suggest that a few states with high expenditure are pulling its average up.

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```

1 #Q2,3. Plot relationship btwn Y and X1 Describe graph + relationship.
  Then,
2 #reproduce with Region and display w/ diff symbols and colors.
3
4 #Step 1: y is numerical, X1 is numerical -> use scatterplot

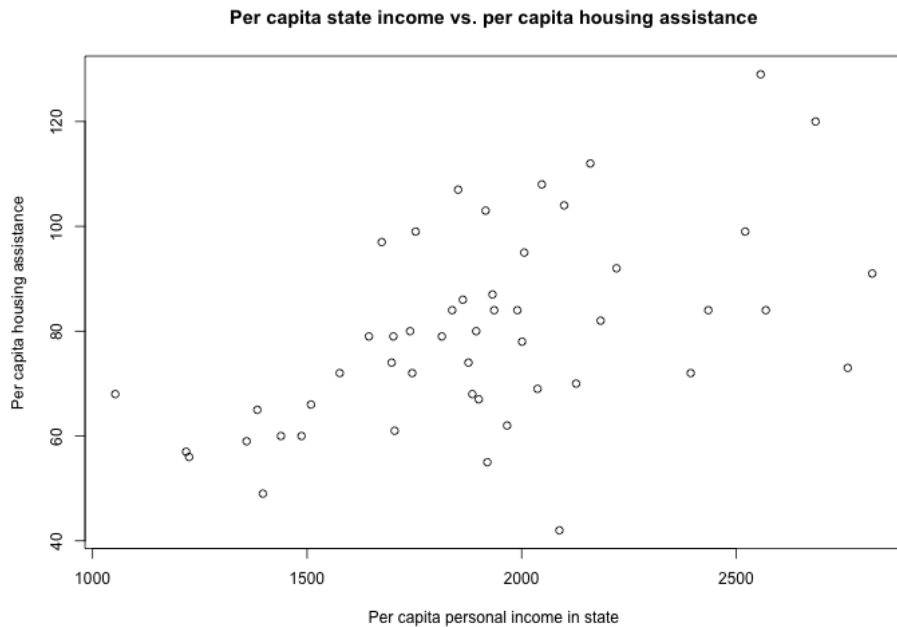
```

```

5 png("graph_y-X1.png", width = 700, height = 500)
6 plot(expenditure$X1, expenditure$Y,
7       main = "Per capita state income vs. per capita housing assistance",
8       xlab = "Per capita personal income in state",
9       ylab = "Per capita housing assistance")
10 dev.off()
11 #The scatterplot shows a moderate positive relationship (r = 0.532), but
12 #are spread out

```

Figure 3: Scatterplot of Y and X1.



Answer: There is a moderate positive relationship between per capita income and housing assistance expenditure.

I use a scatterplot because both variables are numeric. Figure 3 shows that states with higher per capita personal income (X1) generally spend more per capita on housing assistance (Y), with a correlation of $r = 0.532$. However, the points are pretty spread out, so income by itself does not predict spending closely.

```

1 #Step 2: add region with different symbols
2
3 png("graph_y-X1-region.png", width = 700, height = 500)
4 plot(expenditure$X1, expenditure$Y,
5       col = as.numeric(expenditure$Region),
6       pch = as.numeric(expenditure$Region),
7       main = "Housing assistance and income by region",
8       xlab = "Per capita personal income in state",

```

```

9     ylab = "Per capita housing assistance")
10  legend("topleft",
11        legend = levels(expenditure$Region),
12        col = 1:4,
13        pch = 1:4)
14  dev.off()
15  #as.numeric() turns the 4 regions into 1–4, which plot() uses as the
    color and
16  #symbol (pch) codes, so each region gets its own color and symbol
17  #legend() added because plot() doesn't make one automatically
18
19  #The South's states cluster at low income and low spending, and show the
    clearest positive trend. North Central's states are in the middle on
20  #both.
21  #The Northeast mostly has higher incomes, but its spending varies a lot.
22  #The West is the most spread out, with spending from about 42 to 129 at
    similar income levels, so the relationship between X1 and Y looks
23  #different depending on the region.
24

```

Figure 4: Scatterplot of Y and X1, with regions shown by different colors and symbols.

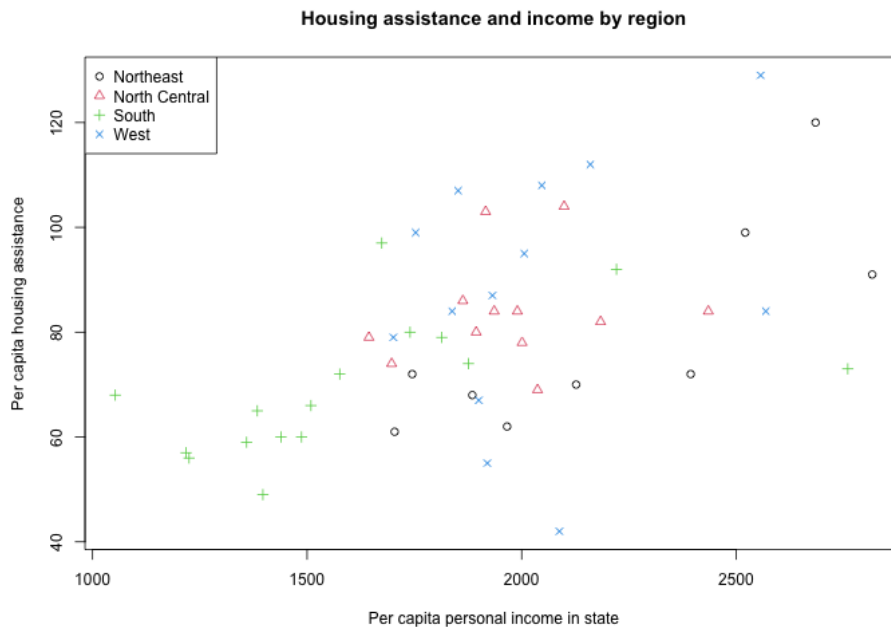


Figure 4 shows the same scatterplot, with each region displayed using a different color and symbol. This reveals that the overall pattern hides differences between regions. The South's states cluster at low income and low spending, and show the clearest positive trend. North Central's states are in the middle on both. The Northeast mostly has higher incomes, but its spending varies a lot. The West is the most spread out, with spending from about 42 to 129 at similar income levels. This suggests the

relationship between X_1 and Y looks different depending on the region.